

Boss Challenge — Paper 30 (Class 7)

Winds, Storms & Cyclones | Nutrition in Animals | Data Handling | Perimeter & Area

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The push that moving and still air presses on everything around it is called air:
(A) speed
(B) volume
(C) pressure
(D) weight
2. The taking in of food and its use by the body to stay alive and grow is called:
(A) nutrition
(B) excretion
(C) respiration
(D) reproduction
3. A collection of numbers gathered to give some useful information is called:
(A) an angle
(B) a graph
(C) data
(D) a ruler
4. The total length of the boundary that goes all the way around a closed figure is called its:
(A) radius
(B) perimeter
(C) area
(D) volume
5. Wind is nothing other than air that is:
(A) frozen
(B) heavy
(C) moving
(D) still
6. The taking of food into the body, the very first step of nutrition, is called:
(A) digestion
(B) ingestion
(C) absorption
(D) egestion
7. The average of a set of numbers, found by adding them all and dividing by how many there are, is called the:
(A) mode
(B) mean
(C) range
(D) median

8. The amount of flat surface that a closed figure covers is called its:
- (A) height
 - (B) area
 - (C) diameter
 - (D) perimeter
9. When air is heated it expands, becomes lighter, and therefore tends to:
- (A) rise up
 - (B) stay still
 - (C) sink down
 - (D) turn solid
10. The breaking down of complex food into simpler substances the body can absorb is called:
- (A) digestion
 - (B) ingestion
 - (C) egestion
 - (D) assimilation
11. Wind speeds on three days were recorded as 20, 30 and 40 km/h. The mean (average) wind speed is:
- (A) 40 km/h
 - (B) 10 km/h
 - (C) 30 km/h
 - (D) 90 km/h
12. For a rectangle whose length is l and whose breadth is b , the perimeter equals:
- (A) $l \times b$
 - (B) $l + b$
 - (C) $2 \times (l + b)$
 - (D) $4 \times l$
13. When warm air rises over a hot region, the air that rushes in to take its place is the cooler air that is:
- (A) frozen solid
 - (B) perfectly still
 - (C) denser and heavier
 - (D) lighter and warmer
14. The removal of the undigested, unwanted part of food from the body is called:
- (A) digestion
 - (B) egestion
 - (C) ingestion
 - (D) absorption

15. The difference between the highest and the lowest value in a set of data is called the:
- (A) median
 - (B) mean
 - (C) range
 - (D) mode
16. The area of a rectangle of length l and breadth b is found using:
- (A) $4 \times l$
 - (B) $2 \times (l + b)$
 - (C) $l + b$
 - (D) $l \times b$
17. Blowing hard over the top of a strip of paper makes it lift, showing that fast-moving air has a:
- (A) higher pressure
 - (B) lower pressure
 - (C) no effect at all
 - (D) greater weight
18. The long tube running from the mouth to the anus, through which food passes, is called the:
- (A) wind pipe
 - (B) alimentary canal
 - (C) blood vessel
 - (D) backbone
19. The value that occurs most often in a set of data is called the:
- (A) mode
 - (B) range
 - (C) median
 - (D) mean
20. The area of a square of side s is given by:
- (A) $s \times s$
 - (B) $2 \times s$
 - (C) $s + s$
 - (D) $4 \times s$
21. On a very large scale, winds are caused mainly by the uneven:
- (A) rising of mountains
 - (B) heating of the Earth
 - (C) spinning of clouds
 - (D) cooling of the oceans
22. The sharp, chisel-shaped front teeth used mainly for cutting and biting food are the:
- (A) incisors
 - (B) canines
 - (C) molars
 - (D) premolars

23. When data is arranged in order, the value that lies exactly in the middle is called the:
- (A) range
 - (B) mode
 - (C) median
 - (D) mean
24. The perimeter of a square of side s is:
- (A) $2 \times s$
 - (B) $s \times s$
 - (C) $4 \times s$
 - (D) $8 \times s$
25. During the daytime, near the coast, the wind that blows from the cool sea towards the warm land is called the:
- (A) monsoon wind
 - (B) sea breeze
 - (C) cyclone
 - (D) land breeze
26. The flat, broad back teeth that crush and grind the food into a paste are the:
- (A) milk teeth
 - (B) canines
 - (C) incisors
 - (D) molars
27. Data shown using bars of equal width, whose heights stand for the amounts, is displayed as a:
- (A) protractor
 - (B) number line
 - (C) bar graph
 - (D) compass
28. The floor of a rectangular cyclone shelter is 8 m long and 5 m broad. Its area is:
- (A) 26 square metres
 - (B) 13 square metres
 - (C) 80 square metres
 - (D) 40 square metres
29. At night near the coast the land cools faster than the sea, so the wind blows from land to sea; this is the:
- (A) storm surge
 - (B) trade wind
 - (C) sea breeze
 - (D) land breeze

- 30.** The watery juice in the mouth that wets the food and begins to break down starch is called:
- (A) bile
 - (B) sweat
 - (C) saliva
 - (D) blood
- 31.** A way of showing data using pictures or symbols, where each symbol stands for a fixed number, is a:
- (A) pictograph
 - (B) range
 - (C) bar graph
 - (D) median
- 32.** A rectangular cattle shed is 10 m long and 6 m wide. The length of fencing needed to go right around it is:
- (A) 60 metres
 - (B) 26 metres
 - (C) 16 metres
 - (D) 32 metres
- 33.** The seasonal winds that reverse their direction with the seasons and bring heavy rain to India are the:
- (A) land breezes
 - (B) tornado winds
 - (C) sea breezes
 - (D) monsoon winds
- 34.** The muscular food pipe that carries swallowed food from the mouth down to the stomach is the:
- (A) liver
 - (B) oesophagus
 - (C) wind pipe
 - (D) small intestine
- 35.** To compare two sets of data side by side on the same chart, the best choice is a:
- (A) double bar graph
 - (B) pie of one slice
 - (C) single bar graph
 - (D) number line
- 36.** A square field has each side 9 m long. Its area is:
- (A) 36 square metres
 - (B) 81 square metres
 - (C) 90 square metres
 - (D) 18 square metres

37. A storm with strong winds, lightning, thunder and heavy rain that forms in hot, humid weather is called a:
- (A) sea breeze
 - (B) drought
 - (C) land breeze
 - (D) thunderstorm
38. Food is pushed along the food pipe and gut by the squeezing, wave-like movement of its muscular walls, called:
- (A) breathing
 - (B) absorption
 - (C) peristalsis
 - (D) digestion
39. The mean of the first five whole numbers 0, 1, 2, 3 and 4 is:
- (A) 5
 - (B) 4
 - (C) 10
 - (D) 2
40. The area of a triangle with base b and height h is given by:
- (A) $\frac{1}{2} \times b \times h$
 - (B) $b + h$
 - (C) $2 \times b \times h$
 - (D) $b \times h$
41. A very strong storm with violent winds whirling around a region of very low pressure is called a:
- (A) rainbow
 - (B) cyclone
 - (C) sea breeze
 - (D) monsoon
42. The J-shaped, bag-like organ that churns food and mixes it with acidic digestive juices is the:
- (A) small intestine
 - (B) liver
 - (C) tongue
 - (D) stomach
43. In a set of marks 7, 9, 9, 9 and 11, the mode is:
- (A) 11
 - (B) 7
 - (C) 45
 - (D) 9

44. Work out the area of a triangle whose base measures 10 cm and whose height is 6 cm:
- (A) 30 square cm
 - (B) 60 square cm
 - (C) 16 square cm
 - (D) 32 square cm
45. The calm region right at the centre of a cyclone, where the winds are light and the sky may clear, is called the:
- (A) eye of the cyclone
 - (B) storm surge
 - (C) edge of the cyclone
 - (D) monsoon trough
46. Most of the digestion is completed and the digested food is absorbed in the longest part of the gut, the:
- (A) food pipe
 - (B) stomach
 - (C) large intestine
 - (D) small intestine
47. Arranged in order, the data 3, 5, 7, 9 and 11 has a median of:
- (A) 7
 - (B) 3
 - (C) 11
 - (D) 5
48. The area of a parallelogram with base b and height h is found using:
- (A) $b + h$
 - (B) $\frac{1}{2} \times b \times h$
 - (C) $b \times h$
 - (D) $4 \times b$
49. Indian cyclones grow over warm sea water; what a cyclone needs to form and grow stronger is:
- (A) ice on high mountains
 - (B) warm, moist air over the sea
 - (C) still air over cities
 - (D) cold, dry desert air
50. Tiny finger-like folds lining the small intestine that greatly increase its surface for absorbing food are called:
- (A) teeth
 - (B) villi
 - (C) ribs
 - (D) nerves

51. Suppose four numbers have a mean of 10; the total you get by adding all four is:
- (A) 2.5
 - (B) 14
 - (C) 40
 - (D) 10
52. A parallelogram has a base of 12 cm and a height of 5 cm. Its area is:
- (A) 60 square cm
 - (B) 17 square cm
 - (C) 30 square cm
 - (D) 34 square cm
53. A dark, funnel-shaped cloud that reaches down from the sky to the ground, whirling very fast, is a:
- (A) sea breeze
 - (B) rainbow
 - (C) tornado
 - (D) monsoon
54. The largest gland in the body, which makes bile to help digest fats, is the:
- (A) heart
 - (B) liver
 - (C) kidney
 - (D) stomach
55. The range of the daily highest temperatures 28° , 31° , 26° and 33° is:
- (A) 26°
 - (B) 7°
 - (C) 33°
 - (D) 59°
56. Area is measured in square units, while perimeter is measured in:
- (A) units of length
 - (B) degrees
 - (C) square units
 - (D) cubic units
57. Wind speed at a weather station is measured using the instrument called the:
- (A) thermometer
 - (B) barometer
 - (C) rain gauge
 - (D) anemometer

58. The watery juice made by the liver that helps in the digestion of fats is called:

- (A) bile
- (B) mucus
- (C) plasma
- (D) saliva

59. Over four health check-ups a cow was found to have 32, 32, 32 and 30 teeth. The mode of these counts is:

- (A) 2
- (B) 30
- (C) 126
- (D) 32

60. To find the cost of fencing all around a rectangular field, you should multiply the rate per metre by the field's:

- (A) height
- (B) perimeter
- (C) area
- (D) diagonal

61. A steady fall in the reading on a barometer often warns that a storm is on its way because the air pressure is:

- (A) staying perfectly fixed
- (B) rising steadily
- (C) dropping
- (D) turning into wind speed

62. The last, wider part of the gut, where extra water is absorbed from the undigested food, is the:

- (A) small intestine
- (B) large intestine
- (C) stomach
- (D) oesophagus

63. When you toss a fair coin once, the chance of getting a head is:

- (A) impossible
- (B) 1 out of 6
- (C) certain
- (D) 1 out of 2

64. To find the cost of laying tiles to cover a rectangular floor, you should multiply the rate per square metre by the floor's:

- (A) perimeter
- (B) length only
- (C) one diagonal
- (D) area

65. Today, the movement and strength of a cyclone far out at sea are mainly tracked using:
- (A) a kitchen thermometer
 - (B) a hand-held compass
 - (C) counting the rainfall by eye
 - (D) satellites and radars
66. The tiny one-celled animal Amoeba captures its food using finger-like extensions of its body called:
- (A) gills
 - (B) teeth
 - (C) pseudopodia
 - (D) roots
67. An event that is sure to happen, such as the Sun rising in the east, has a chance that is:
- (A) certain
 - (B) equally likely to a coin head
 - (C) never
 - (D) impossible
68. A square and a rectangle each have a perimeter of 24 cm. The square has side 6 cm; the rectangle is 8 cm by 4 cm. The one with the larger area is the:
- (A) neither has any area
 - (B) square
 - (C) rectangle
 - (D) they are equal
69. When a cyclone warning is issued, the safest thing for people in the area to do is to:
- (A) go out to watch the storm
 - (B) ignore the warning
 - (C) move to a safe shelter
 - (D) stand under a tall tree
70. Inside an Amoeba, the trapped food is digested within a small bubble-like space called the:
- (A) villus
 - (B) nucleus
 - (C) stomach
 - (D) food vacuole
71. An event that can never happen, such as a tossed coin landing on its thin edge and standing forever, is described as:
- (A) certain
 - (B) 1 out of 2
 - (C) very likely
 - (D) impossible

72. In a circle, the distance right across through the centre is the diameter, and it is always the radius:

- (A) doubled
- (B) halved
- (C) the same length
- (D) squared

73. The sudden rise of sea water that floods low coastal land when a cyclone strikes is called a storm:

- (A) shadow
- (B) surge
- (C) breeze
- (D) shower

74. Grass-eating animals like cows quickly swallow grass and store it in a part of the stomach called the:

- (A) rumen
- (B) villus
- (C) liver
- (D) food vacuole

75. To find the median of a list of numbers, the first thing you must always do is:

- (A) multiply them
- (B) pick the largest
- (C) arrange them in order
- (D) add them all up

76. The distance all the way around the edge of a circle is given its own special name, the:

- (A) circumference
- (B) diameter
- (C) radius
- (D) area

77. The roof of a poorly built house can be blown off in a cyclone because the very fast wind above it creates:

- (A) a vacuum with no air at all
- (B) extra weight pressing down
- (C) lower pressure above the roof
- (D) higher pressure above the roof

78. Cattle and other grass-eaters later bring the stored, partly digested grass back to the mouth to chew slowly; this is called chewing the:

- (A) cud
- (B) mucus
- (C) saliva
- (D) bile

79. Cyclones struck a coast in the months March, July, July and November over four years. The mode of these months is:

- (A) four
- (B) November
- (C) March
- (D) July

80. A rectangular plot is 7 m long and 3 m wide. Its perimeter is:

- (A) 10 metres
- (B) 14 metres
- (C) 21 metres
- (D) 20 metres

81. To reduce the loss of life in cyclone-prone coastal areas, governments build strong cyclone:

- (A) swimming pools
- (B) shelters
- (C) kite grounds
- (D) glass towers

82. Grass is rich in a tough material that grass-eaters digest with the help of bacteria living in their gut; this material is:

- (A) saliva
- (B) bile
- (C) starch
- (D) cellulose

83. If one very large value is added to a small set of numbers, the measure most pulled upward by it is the:

- (A) mode
- (B) median
- (C) range
- (D) mean

84. One square metre is the same as the number of square centimetres equal to:

- (A) 1000
- (B) 10
- (C) 100
- (D) 10000

85. When a cyclone is likely within about 48 hours, the weather office issues a cyclone:

- (A) holiday
- (B) festival
- (C) alert
- (D) rainbow

86. The opening at the end of the alimentary canal through which the undigested waste leaves the body is the:

- (A) mouth
- (B) nose
- (C) anus
- (D) ear

87. The mean of the four numbers 6, 8, 10 and 12 is:

- (A) 9
- (B) 12
- (C) 36
- (D) 4

88. Doubling both the length and the breadth of a rectangle makes its area become:

- (A) the same
- (B) half as large
- (C) twice as large
- (D) four times as large

89. During a cyclone, people should trust official news and warnings and should never:

- (A) believe rumours
- (B) move to a shelter
- (C) listen to the weather office
- (D) keep emergency supplies

90. The using up of absorbed, digested food by body cells to build the body and release energy is called:

- (A) ingestion
- (B) assimilation
- (C) digestion
- (D) egestion

91. A pictograph shows that one picture of a tree stands for 10 trees. A row of 4 such pictures means:

- (A) 14 trees
- (B) 40 trees
- (C) 10 trees
- (D) 4 trees

92. A footpath of bricks runs around the inside edge of a square park of side 20 m. The length of the path's outer boundary equals the park's:

- (A) perimeter
- (B) diagonal
- (C) side only
- (D) area

- 93.** Planting rows of trees along a coast helps in a cyclone because the trees act as a:
- (A) roof over the whole town
 - (B) magnet that pulls the storm in
 - (C) windbreak that slows the wind
 - (D) source of more wind
- 94.** Adding a few drops of iodine to food and seeing it turn blue-black tells us the food contains:
- (A) water
 - (B) protein
 - (C) starch
 - (D) fat
- 95.** On a double bar graph comparing rainfall in two towns, each town is shown for every month by:
- (A) no bars at all
 - (B) two bars side by side
 - (C) a single bar
 - (D) a circle
- 96.** A square cyclone-shelter floor and a rectangular one both cover the same area of 36 square metres. The square one has side:
- (A) 9 metres
 - (B) 18 metres
 - (C) 6 metres
 - (D) 36 metres
- 97.** A spinning storm of the kind called a cyclone in India is given the name 'hurricane' in:
- (A) America
 - (B) the deep desert
 - (C) India
 - (D) outer space
- 98.** If you chew a piece of bread (which contains starch) for a long time, it begins to taste slightly sweet because saliva turns the starch into:
- (A) protein
 - (B) sugar
 - (C) fat
 - (D) salt
- 99.** Counting how many students scored each mark and writing the counts in a neat table makes the data easier to:
- (A) read and understand
 - (B) turn into a wind
 - (C) make impossible to use
 - (D) hide from everyone

100. A rectangle and a square both have an area of 36 square metres, but their perimeters differ; this shows that the same area can go with a:

- (A) zero perimeter
- (B) equal perimeter always
- (C) the same shape always
- (D) different perimeter